

BIOL90042 tutorial 6

- In this tutorial, we will practice questions of the type you can expect on your mid-semester test next week.
- Overall, the aim of questions is to test your broader understanding of the topics covered in this subject, rather than memorisation of concepts
- You may use the summary sheets provided in the coding workshops to help you, but nothing else.

Multiple choice questions

These are all worth one hypothetical mark, unless otherwise indicated

Question 1

When should I run code in RStudio's console rather than in an R script or Quarto file?

- A. When the code is very important for my analysis
- B. When the code will take a long time to run
- C. When the code is just to quickly check something, and I don't need to keep a record of it
- D. When the code needs to be shared with others

Question 2

I'm trying to read in a file in R, which is called 'ELISA_data.csv' and located in the 'experiments' folder on my desktop. Which of the following `read_csv()` calls is correct?

- A. `read_csv("~/Desktop/experiments/")`
- B. `read_csv("~/Desktop/experiments/elisa_data.csv")`
- C. `read_csv("~/Desktop/experiments/ELISA_data.csv")`

D. `read_csv("~/Desktop/experiments/ELISA_data.csv")`

Question 3 [2 marks]

Fill in the blanks to complete the following paragraph about good coding practice and organisation:

“When doing data analysis, it is very important to stay organised. We can organise all of our scripts and data using a [1], and ensure that our file names are [2]. While writing the actual code, we can use [3] to describe our thought process, and finally share our code with others via [4].”

1. folder / quarto file / R project /
2. detailed / as short as possible / include spaces
3. variables / functions / comments
4. email / git / GitHub

Question 4

What is the difference between the git commands ‘add’ and ‘commit’?

- A. Nothing, `git add` and `git commit` do the exact same thing
- B. `git add` is used to save your files, and `git commit` is used to download them
- C. `git add` saves your changes locally, while `git commit` uploads them to GitHub
- D. `git add` is used to add changes to git’s tracking, while `git commit` saves your changes to the local repository

Question 5

What is quarto?

- A. A way we can combine code and text easily
- B. The same thing as an R script
- C. A new program for running R code
- D. A way to keep track of changes to our code

Question 6

A scatterplot would **not** be an appropriate visualisation for which pair of variables?

- A. Mouse tail length and whisker length
- B. Survival time and cancer stage (stage 1, 2, 3 and 4)

- C. Drug dosage and gene expression
- D. B cell count and T cell count

Question 7

Which of the following is **not** something we should consider when picking colours for a graph?

- A. Whether the color scheme matches the type of data being shown (e.g., sequential for ordered values, qualitative for categories)
- B. That the intensity of the colours are consistent
- C. Making the sure the colours match the default R colour palette
- D. Making sure it is colourblind-safe

Question 8

What are facets, in the context of data visualisation?

- A. A colour palette
- B. A special type of plot
- C. When you colour the points by a column in your dataframe
- D. A way to create several 'mini plots' based on a categorical variable

Question 9

Which of the following is **not** a reason we use replicates in RNA-seq experiments?

- A. Because they increase the sequencing depth of each sample
- B. Because they increase statistical power
- C. Because they allow us to capture biological variation
- D. Because they make us more confident in our results

Question 10 [2 marks]

Fill in the blanks to complete the following paragraph about biological data types:

"There are many different technologies that can be used to generate biological data. For example, if we are interested in knowing the sequence of DNA, we can use [1]. Many types of biological data require specialised analysis techniques, like [2] for transcriptomic and proteomic data. After this, we are generally interested in finding things that are [3], which we can visualise on a [4]."

1. NGS / mass spectrometry / microscopy
2. squaring / log-transformation / pivoting
3. differentially expressed / very high / very low
4. bar chart / boxplot / volcano plot

Question 11

Which of the following is an advantage of using multiple linear regression over simple linear regression?

- A. Because it removes the need to check any model assumptions
- B. Because it guarantees the relationship between variables is causal
- C. Because it allows us to examine the effect of multiple variables on an outcome
- D. Because it makes our correlation higher

Question 12

Why is it important to correct for batch effects in RNA-seq?

- A. Because batch effects always increase the total number of genes detected
- B. Because batch effects can be mistaken for real biological differences, leading to false conclusions
- C. Because correcting for batch effects removes the need for biological replicates
- D. Because we want all of our data to be as similar as possible

Question 13

Which of the following is **not** a way we can assess the validity our linear regression model?

- A. By looking at p-values for the model coefficients
- B. By checking residual plots for patterns (to assess linearity and homoscedasticity)
- C. By examining the R^2 value to see how much variance is explained
- D. By checking whether there is a positive correlation between the dependent and independent variables

Question 14

What is the atomic type of the value "TRUE" ?

- A. Data frame

- B. Logical (boolean)
- C. Character
- D. Numeric

Question 15

What would be the result of running the following code: `summary(iris$Sepal.Length)`

- A. An error
- B. Various statistics about the `iris` dataframe
- C. The mean of the `Sepal.Length` column in the `iris` dataframe
- D. Various statistics about the `Sepal.Length` column in the `iris` dataframe

Question 16

Which of the following would sort the `temp` column of `beaver1` in descending order?

- A. `beaver1 %>% arrange(temp)`
- B. `beaver1 %>% arrange(desc(temp))`
- C. `beaver1 %>% sort(temp)`
- D. `beaver1 %>% slice_max(temp, n = 5)`

Question 17

I'm trying to add a new column to `beaver1` that contains the temperature in Farenheit. I know the formula for this is $^{\circ}F = (^{\circ}C \times 9/5) + 32$, so I have written the following code:

```
beaver1 %>% mutate(temp_f == (temp * 9/5) + 32)
```

However, I get an error: `! object 'temp_f' not found`

What is the problem?

- A. There is a missing bracket
- B. `temp_f` doesn't exist as a column, it should just be `temp`
- C. The use of `==` inside the `mutate()` function
- D. You can't use the `mutate()` function for this

Question 18

Which of the following would **not** allow me to join two dataframes `df1` and `df2`, by a shared column `col1`?

- A. `left_join(df1, df2, col1)`
- B. `left_join(df1, df2)`
- C. `df1 %>% left_join(df2)`
- D. `left_join(df1, df2, by = "col1")`

Question 19

In the `pivot_longer()` function, the `names_to` argument refers to:

- A. The names of columns to pivot
- B. The name of a new column which will contains the pivoted values
- C. The name of the dataframe that you want to pivot into
- D. The name of a new column which will contain the names of the columns you pivoted

Question 20

In the `ggplot2` package, what is the `aes()` function used for?

- A. To assign columns from a dataframe to various aesthetics of a plot, like the x and y axes, or colour
- B. To add points to a plot
- C. To create the initial setup of the plot
- D. To describe purely aesthetic features of the plot, like the font size or whether something should be bold

Question 21

I'm trying to make a boxplot, of the `temp` column in `beaver1`, but I got an error. This is my code:

```
beaver1 %>% ggplot(x = temp) + geom_boxplot()
```

And this is the error message: `! stat_boxplot() requires an x or y aesthetic.`

What is wrong?

- A. The function is called `ggplot2()` not `ggplot()`
- B. `x = temp` should be inside the `aes()` function
- C. You can't use the pipe `%>%` with `ggplot2`
- D. To make a boxplot you should use the `boxplot()` function, not `geom_boxplot()`

Question 22

Describe in words what the following code does:

```
airquality %>%  
  replace_na(list(Ozone = 0)) %>%  
  ggplot(aes(x = Ozone, y = Temp)) +  
    geom_point()
```

- A. Nothing, this code would give an error
- B. Removes NA in the ozone column, then makes a scatterplot
- C. Creates a scatterplot of ozone and temperature, after replacing missing values in the ozone column with 0
- D. Replaces all missing values with 0 and then makes a scatterplot of ozone and temperature

Question 23

Why is it important for data to be tidy?

- A. Because it makes our code run faster
- B. Because it allows us to consistently re-use a set of functions across many datasets, which are known as the tidyverse
- C. Because it removes outliers and makes our analysis look cleaner
- D. Because tidyverse functions only work on tidy data

Short-answer questions

Question 1 [2 marks]

I've written the following R code:

```
rnaseq_data %>%  
  summarise(  
    min_expression = min(expression),  
    .by = c(group, gene)  
  )
```

Explain in words what it does. Can you think of another function that could be used to do the same thing?

Question 2 [4 marks]

I've written the following R code, but it contains some errors

```
dosage %>%
  filter(mouse_strain = "CD-1") %>%
  select(id_num, weight_lost_g, cage_number)
  summarise(
    mean_weight_lost = mean(weight_lost_g),
    by = cage_number %>%
  pivot_wider(
    names_from = cage_number,
    values_to = mean_weight_lost)
```

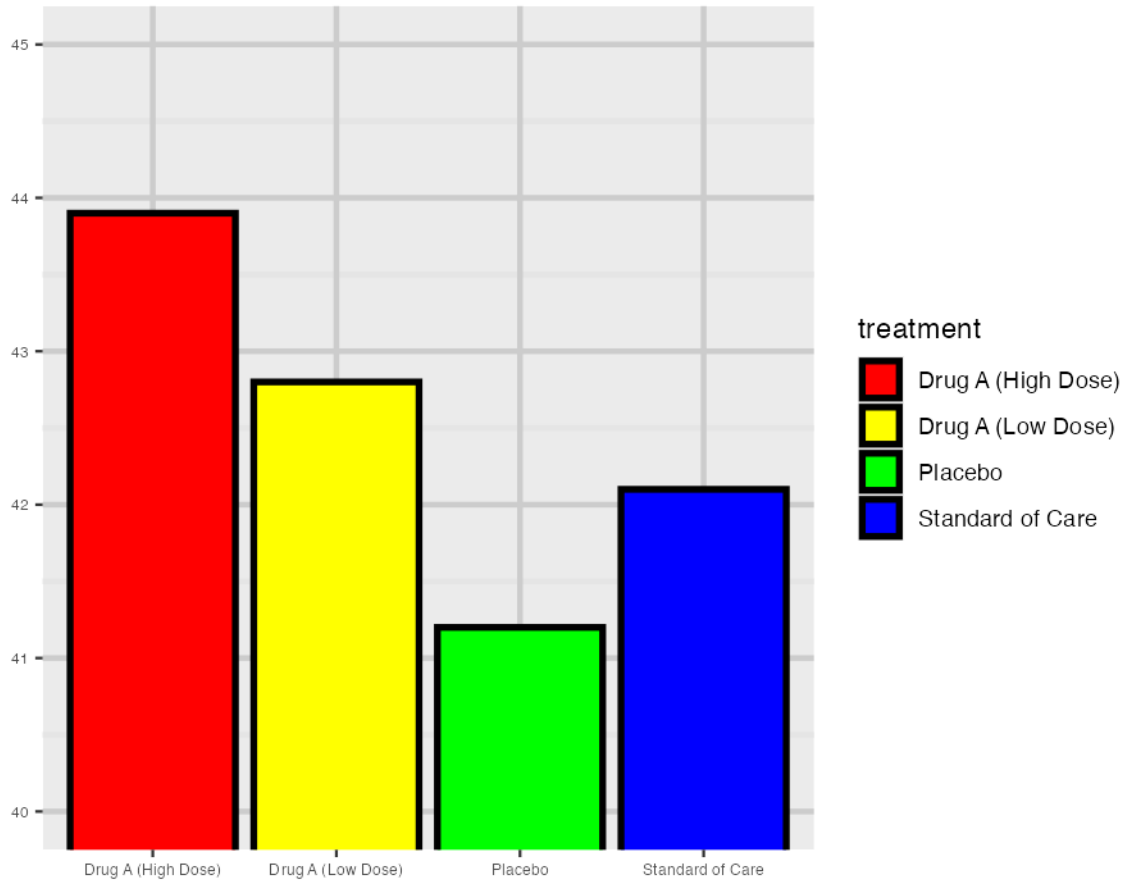
```
Error in parse(text = input): <text>:11:0: unexpected end of input
9:     values_to = mean_weight_lost)
10:
   ^
```

Identify **four** errors in the code (there may be more)

Question 3 [4 marks]

My colleague has made this interesting graph:

NEW DRUG A SHOWS BREAKTHROUGH EFFICACY



Do you think this plot is easy to interpret? Justify your answer by identifying **three** elements of the plot that violate good data visualisation principles.